INCH-POUND

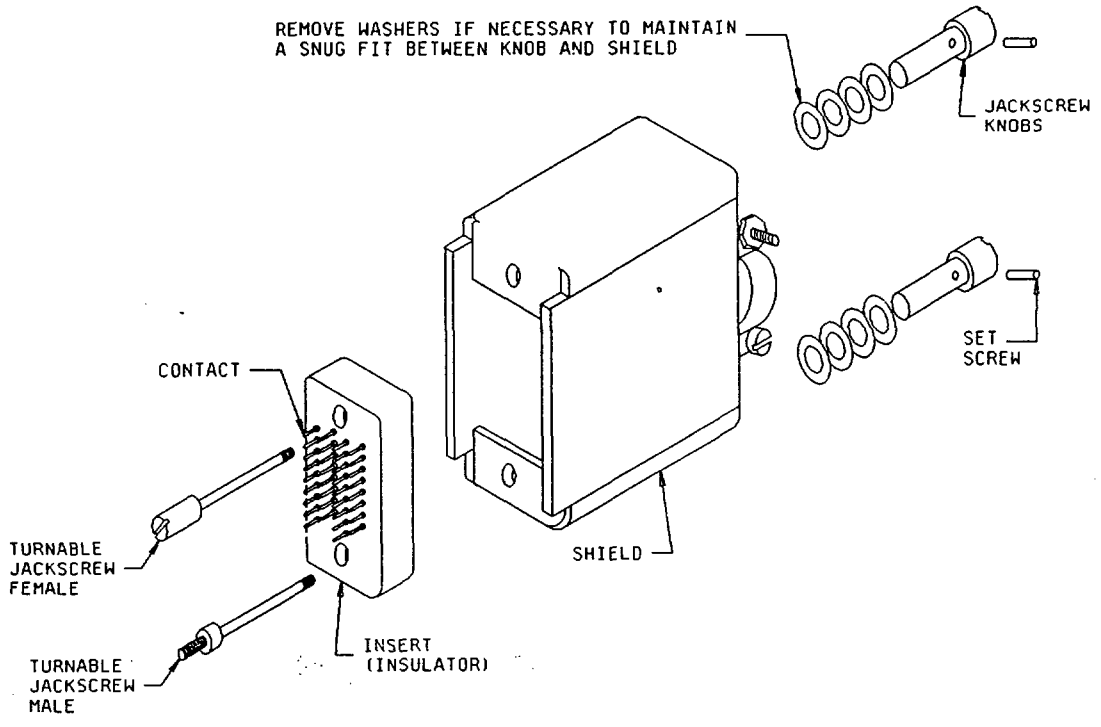
MIL-C-28748/7E
1 April 1993
SUPERSEDING
MIL-C-28748/7D
27 July 1990

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR,
RACK AND PANEL, SOLDER TYPE PIN CONTACTS, SIZE 22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-28748.



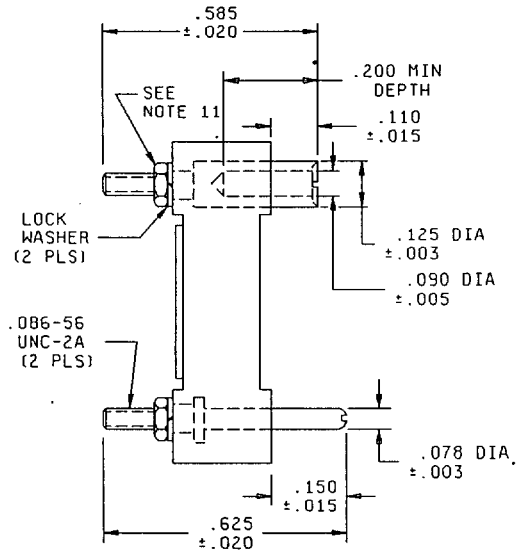
Connector

Configuration A

FIGURE 1. Connector with shields, jackscrews, and guidepins.

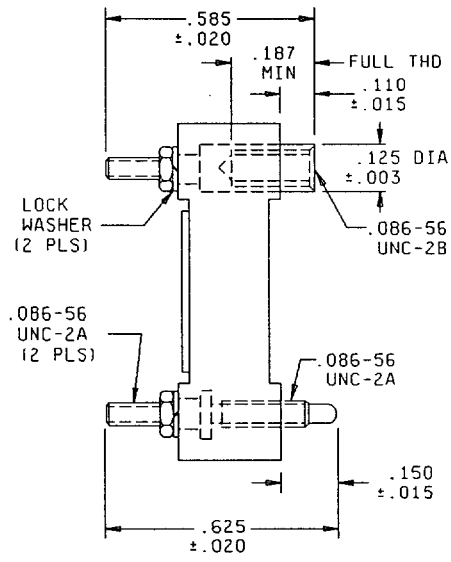
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Connector with guidepins

Configuration B

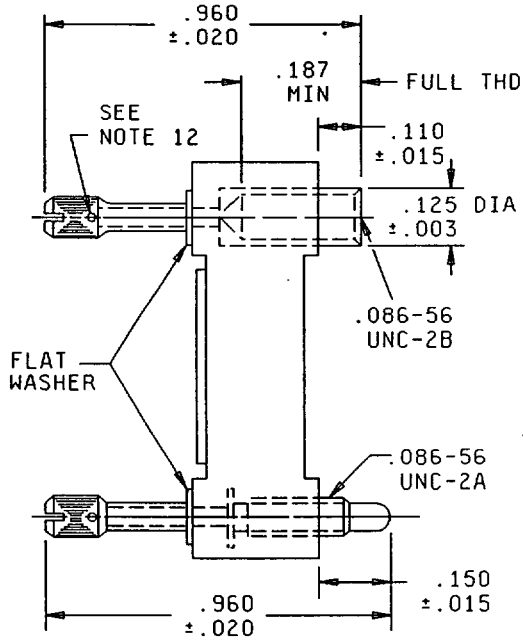


Connector with fixed jackscrews

Configuration C

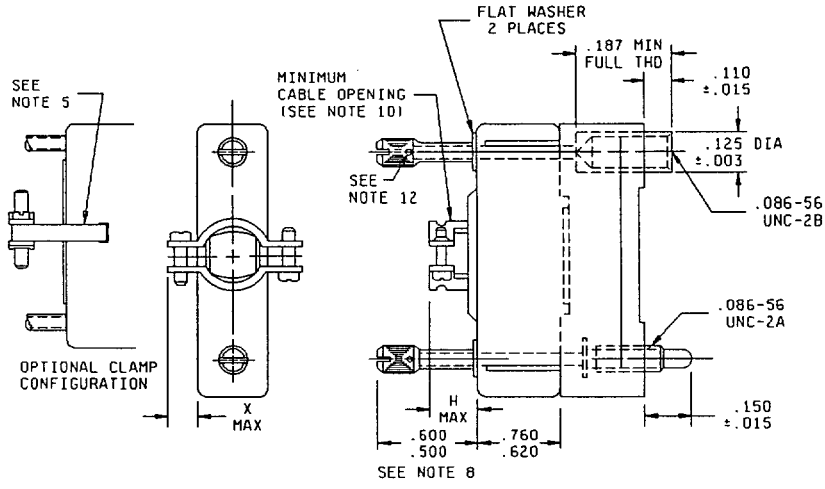
FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Connector with short turnable jackscrews

Configuration D



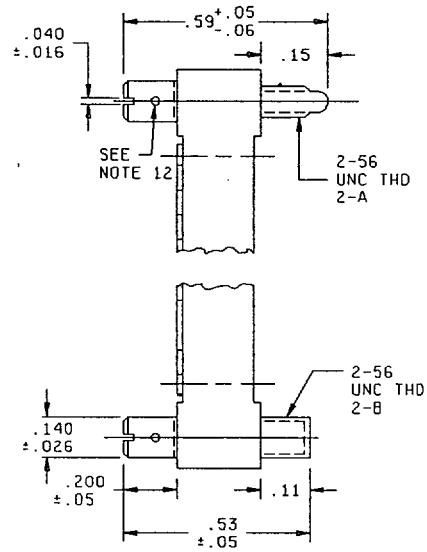
Contact position	X max	H max
4	.210	.300
7	.210	.300
14	.210	.350
20	.210	.350
26	.150	.400
34	.100	.400
44	.100	.500
50	.100	.500

Connector with long turnable jackscrews and shield

Configuration E

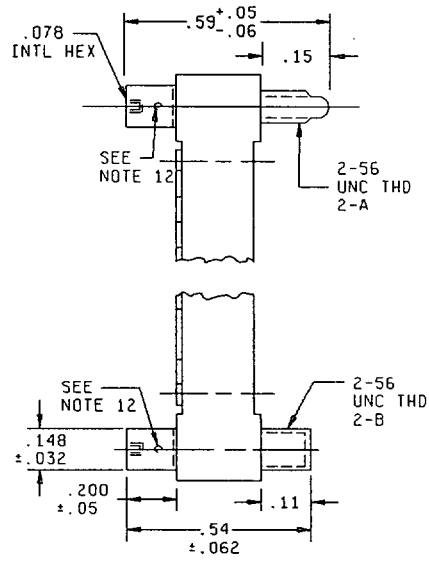
FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Connector with miniature slotted turnable jackscrews

Configuration F



Connector with miniature hexhead turnable jackscrews

Configuration G

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:14Z
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Inches	mm	Inches	mm	Inches	mm
.003	0.08	.110	2.79	.59	15.0
.015	0.38	.125	3.18	.600	15.24
.016	0.41	.140	3.56	.620	15.75
.020	0.51	.148	3.76	.625	15.88
.026	0.66	.15	3.8	.760	19.30
.032	0.81	.150	3.81	.960	24.38
.040	1.02	.187	4.75		
.05	1.30	.200	5.08		
.06	1.50	.500	12.70		
.062	1.57	.53	13.5		
.078	1.98	.54	13.72		
.090	2.29	.585	14.86		

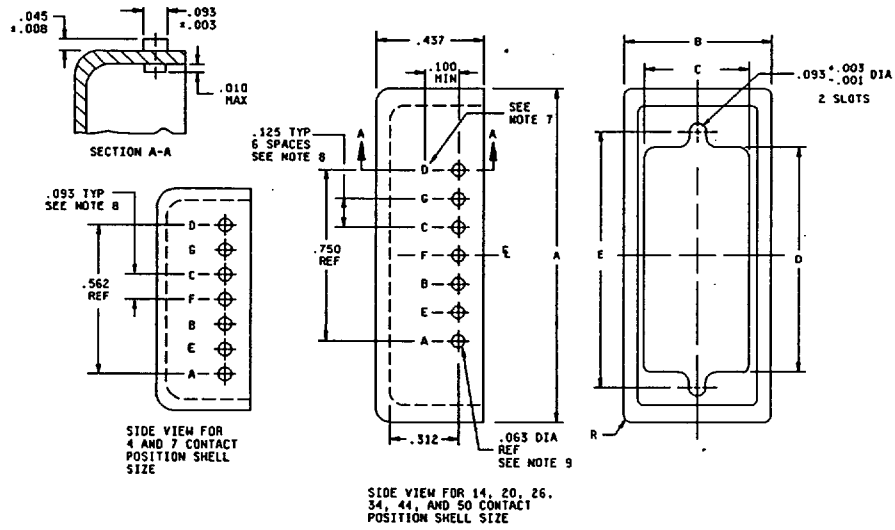
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 inch (0.5 mm) for two place decimals and ±.015 inch (0.38 mm) for three place decimals.
4. Dimensions and configuration shall be in accordance with the above figure.
5. Cable clamp configuration has optional mounting method.
6. Materials for jackscrews and guidepins: Corrosion resistance steel in accordance with QQ-S-763 or ASTM-A-582, class 302, 303, 3038E, 304, or 316, except knobs, washers, and spacers shall be either aluminum alloy or corrosion resistance steel. Finish for washers and spacers shall be passivated or anodized.
7. No side clamp available with insert arrangements A, B, D, E, and F.
8. .730 inch (18.54 mm) maximum for 14-contact connector with side opening shield. .630 inch (16.00 mm) maximum for 50-contact connector.
9. 1.200 inch (30.48 mm) maximum for 44-contact arrangement. 1.520 inch (38.61 mm) maximum for 50-contact arrangement.
10.

Number of contacts	Minimum cable opening
4	.027 (0.69 mm²) in²
7	.027 (0.69 mm²) in²
14	.070 (1.80 mm²) in²
20	.081 (2.06 mm²) in²
26	.089 (2.26 mm²) in²
34	.110 (2.79 mm²) in²
44	.160 (4.06 mm²) in²
50	.170 (4.32 mm²) in²

11. Flat washers are required between lockwashers and insert material.
12. For jackscrew assemblies (excluding one piece constructions) setscrew or groove pin shall be used. Setscrew requires a torque of 2 inch-pound minimum. No parts of the connector assembly shall be permanently displaced from their original, normal, fitted position at the completion of the specified tests.

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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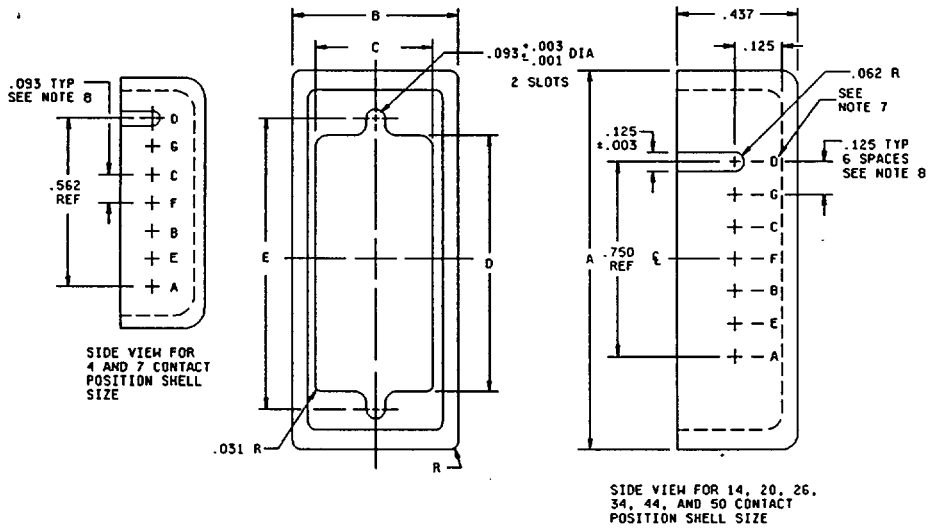
Insert designator	Connector size (number of contacts	A max	B max	C min	D min	E ±.005 (0.13)	R
N	4	.870 (22.10)	.300 (7.62)	.230 (5.84)	.430 (10.92)	.562 (14.27)	.031 (0.79)
P	7	.870 (22.10)	.300 (7.62)	.230 (5.84)	.430 (10.92)	.562 (14.27)	.031 (0.79
Q	14	.970 (24.64)	.370 (9.40)	.300 (5.84)	.530 (13.46)	.625 (15.88)	.062 (1.57)
R	20	1.160 (29.46)	.370 (9.40)	.300 (5.84)	.730 (18.54)	.814 (20.68)	.062 (1.57)
S	26	1.345 (34.16)	.370 (9.40)	.300 (5.84)	.910 (23.11)	1.000 (25.40)	.062 (1.57)
T	34	1.340 (34.04)	.480 (12.19)	.410 (10.41)	.900 (22.86)	1.032 (26.21)	.062 (1.57)
U	44	1.590 (40.39)	.480 (12.19)	.410 (10.41)	1.140 (28.96)	1.281 (32.54)	.062 (1.57)
V	50	1.710 (43.43)	.480 (12.19)	.410 (10.41)	1.270 (32.26)	1.408 (35.76)	.062 (1.57)

Plug shell

Configuration A

FIGURE 2. Shell configurations.

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Insert designator	Connector size (number of contacts	A max	B max	C min	D min	E ±.005 (0.13)	R
N	4	.875 (22.22)	.305 (7.75)	.230 (5.84)	.430 (10.92)	.562 (14.27)	.031 (0.79)
P	7	.875 (22.22)	.305 (7.75)	.230 (5.84)	.430 (10.92)	.562 (14.27)	.031 (0.79
Q	14	.975 (24.76)	.375 (9.52)	.300 (5.84)	.530 (13.46)	.625 (15.88)	.062 (1.57)
R	20	1.165 (29.59)	.375 (9.52)	.300 (5.84)	.730 (18.54)	.814 (20.68)	.062 (1.57)
S	26	1.350 (34.29)	.375 (9.52)	.300 (5.84)	.910 (23.11)	1.000 (25.40)	.062 (1.57)
T	34	1.344 (34.16)	.480 (12.19)	.410 (10.41)	.900 (22.86)	1.032 (26.21)	.062 (1.57)
U	44	1.595 (40.51)	.480 (12.19)	.410 (10.41)	1.140 (28.96)	1.281 (32.54)	.062 (1.57)
V	50	1.715 (43.56)	.480 (12.19)	.410 (10.41)	1.270 (32.26)	1.408 (35.76)	.062 (1.57)

Receptacle shell

Configuration B

FIGURE 2. Shell configurations - Continued.

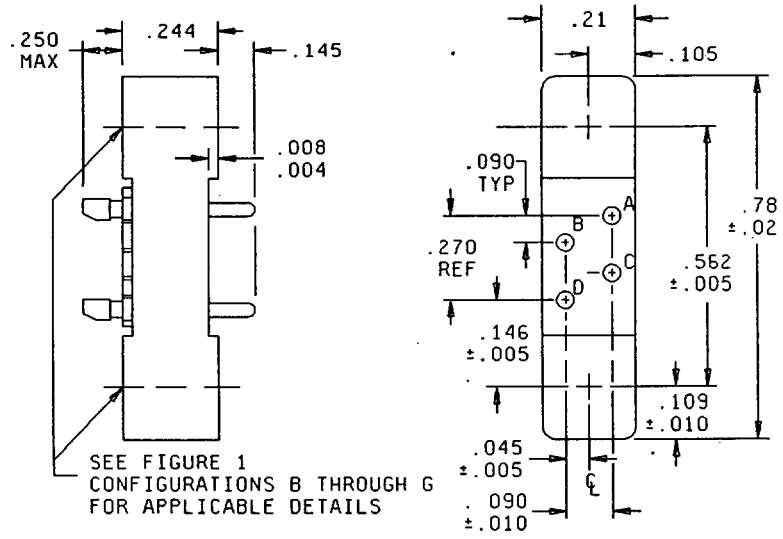
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:14Z
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Inches	mm
.001	0.03
.003	0.08
.008	0.20
.010	0.25
.031	0.79
.045	1.14
.062	1.57
.063	1.60
.093	2.36
.100	2.54
.125	3.18
.312	7.92
.437	11.10
.562	14.27
.750	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Unless otherwise specified, tolerance is ±.016 inch (0.41 mm).
4. Metric equivalents are in parentheses.
5. Material: Aluminum alloy in accordance with QQ-A-250/18, temper 0, or QQ-A-250/11, temper H12; .025 inch (0.64 mm) thick.
6. Finish: Anodize in accordance with MIL-A-8625 type I or II.
7. Polarization position letter is .06 inch (1.5 mm) height minimum as shown using permanent marking slot and letter for applicable location only.
8. The location may vary ±.005 inch (0.13 mm) from the centerline.
9. One polarization pin is used when applicable.
10. Insert designators are informational and used only when ordering a shield or shell as a piece part. Example: M28748/7-ROL01L is a receptacle shell with C polarization for a 20-contact position insert.
11. The polarization feature (pin or groove) shall be adjacent to the side closest to the A contact position.

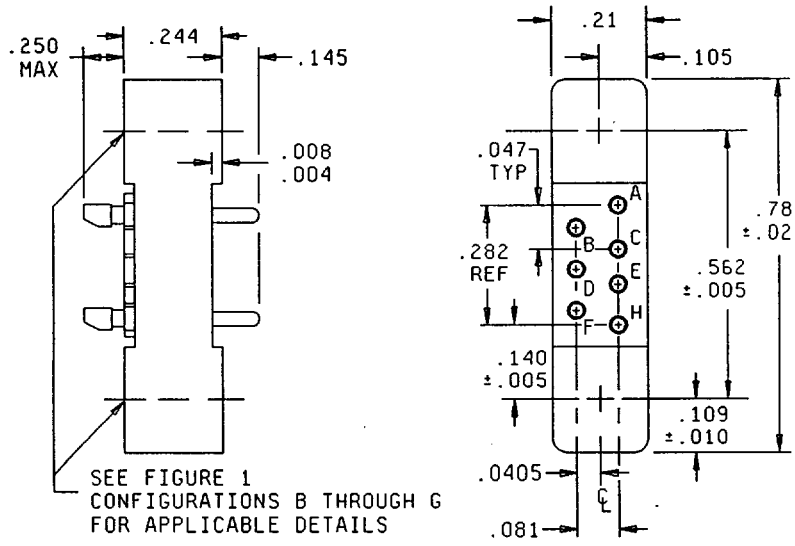
FIGURE 2. Shell configurations - Continued.

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Insert with 4 contacts

Configuration A

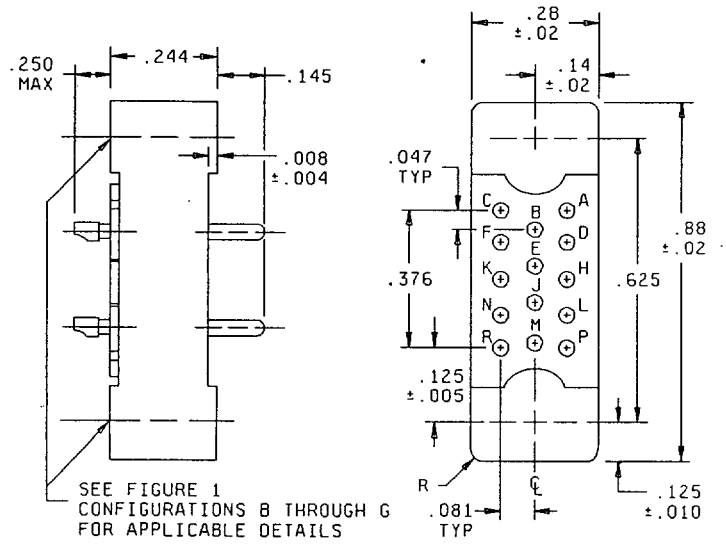


Insert with 7 contacts

Configuration B

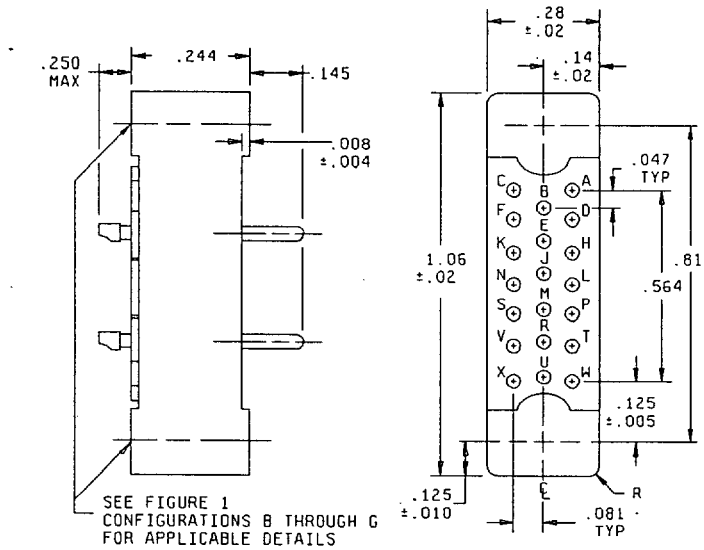
FIGURE 3. Insert arrangements.

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Insert with 14 contacts

Configuration C

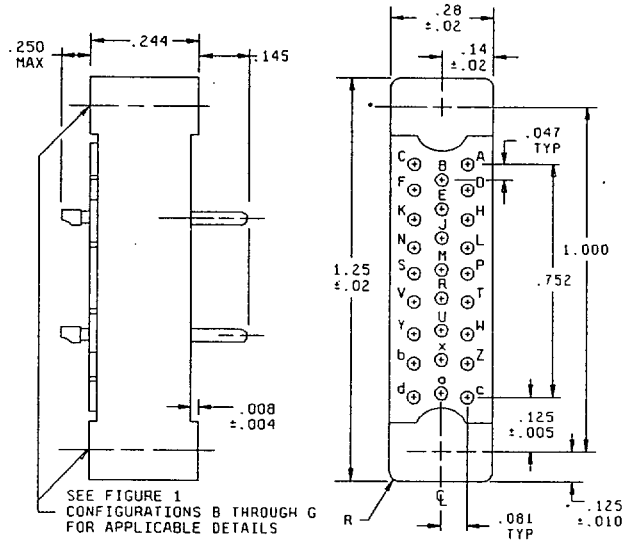


Insert with 20 contacts

Configuration D

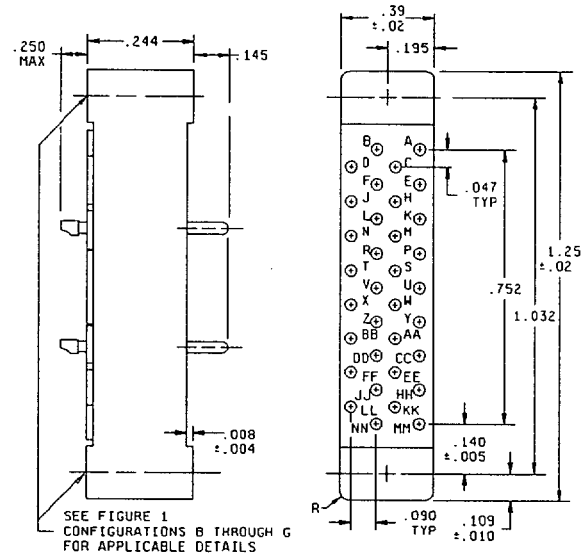
FIGURE 3. Insert arrangements - Continued.

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Insert with 26 contacts

Configuration E

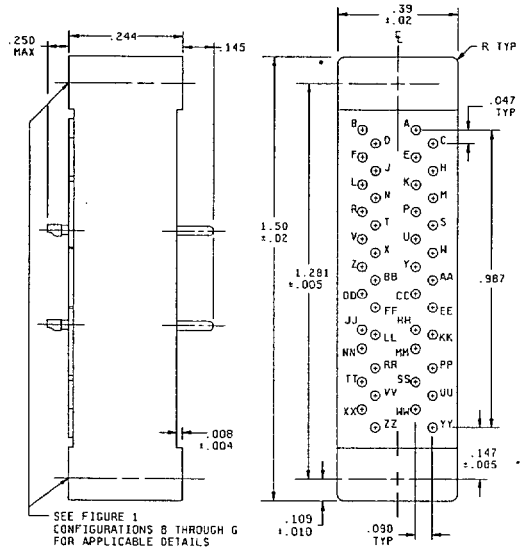


Insert with 34 contacts

Configuration F

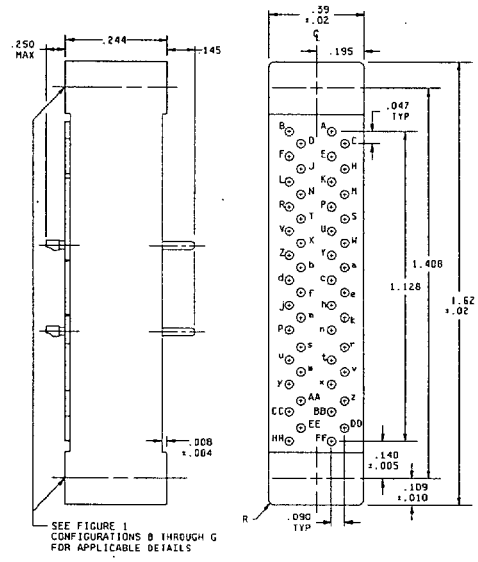
FIGURE 3. Insert arrangements.

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Insert with 44 contacts

Configuration G



Insert with 50 contacts

Configuration H

FIGURE 3. Insert arrangements - Continued.

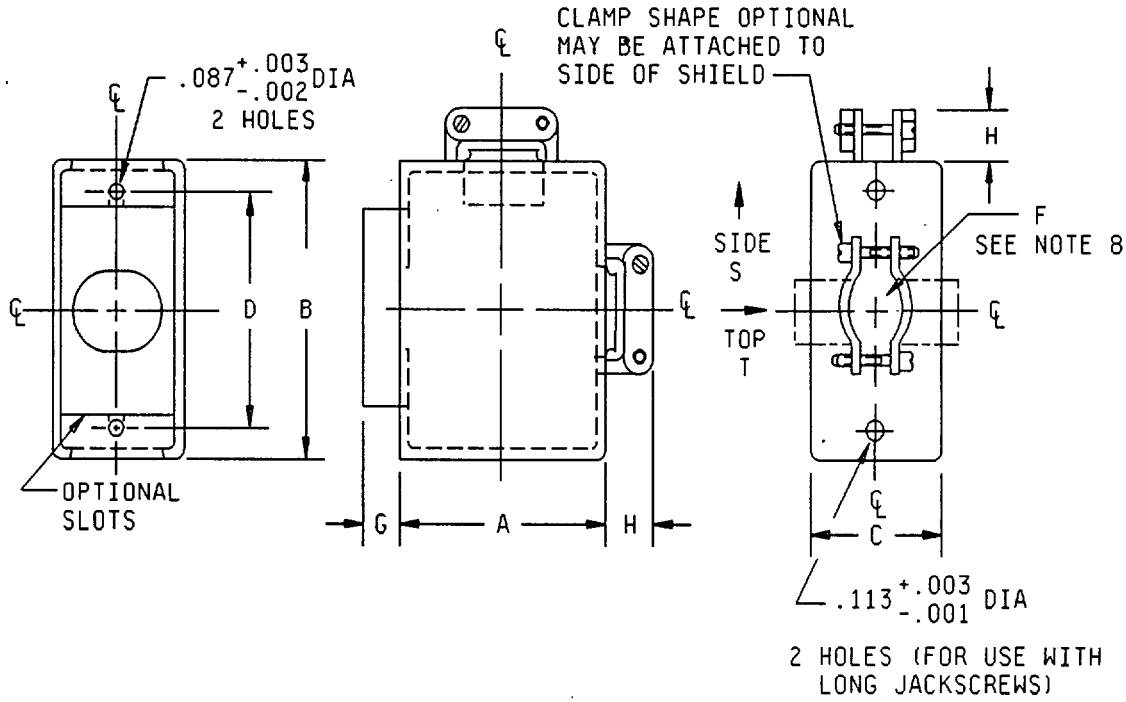
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:14Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm
.004	0.10	.145	3.68	.625	15.88
.005	0.13	.146	3.71	.752	19.10
.008	0.20	.147	3.73	.78	19.18
.010	0.25	.195	4.95	.814	20.68
.02	0.5	.21	5.3	.88	22.4
.0405	1.029	.244	6.20	.987	25.07
.045	1.14	.250	6.35	1.000	25.40
.047	1.19	.270	6.86	1.032	26.21
.081	2.06	.28	7.1	1.06	26.9
.090	2.29	.282	7.16	1.128	28.65
.105	2.67	.350	8.89	1.25	31.8
.109	2.77	.376	9.55	1.281	32.54
.125	3.18	.39	9.9	1.408	35.76
.140	3.56	.562	19.27	1.50	38.1
.14	3.6	.564	14.33	1.62	41.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 inch (0.5 mm) for two place decimals and ±.015 inch (0.38 mm) for three place decimals.
4. Dimensions and configuration shall be in accordance with the above figure.
5. Tolerance between any two adjacent contact centers shall be ±.004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ±.006 inch (0.15 mm).
6. Contact letter identification may be located on either side of the contact hole. All letters shall appear on the front and rear face of each insert.
7. Inserts shall be used with guide pin assemblies or jackscrew assemblies.
8. Insert material: Shall be in accordance with MIL-C-28748.

FIGURE 3. Insert arrangements - Continued.

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.087	2.21
.113	2.87

FIGURE 4. Shield.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:14Z
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Insert designator	Cable opening	Connector size	A max min	B	C	D ±.015 (0.38)	F min cable opening	G ±.05 (1.3)	H max
N	1	4	.176 (19.30) .620 (15.75	.780 (19.81)	.268 (6.81)	.562 (14.27)	.027 (0.69)	.20 (5.1)	.300 (7.62)
P	1	7	.760 (19.30) .620 (15.75)	.780 (19.81)	.268 (6.81)	.562 (14.27)	.027 (0.69)	.20 (5.1)	.300 (7.62)
Q	1 or 2	14	.760 (19.30) .620 (17.75)	.875 (22.22)	.348 (8.84) .334 (8.48) ±.030 (0.76)	.625 (15.88)	.07 (1.8) .037 (0.94)	.20 (5.1)	.350 (8.89
R	1	20	.730 (19.30) .620 (15.75)	1.064 (27.03)	.348 (8.84)	.814 (20.68)	.081 (2.06)	.215 (5.46)	.350 (8.89)
S	1 or 2	26	.730 (19.30) .620 (15.75)	1.250 (31.75)	.348 (8.84)	1.000 (25.40)	.089 (2.26	.20 (5.1)	.400 (10.16)
T	1	34	.730 (19.30) .620 (15.75)	1.250 (31.75)	.452 (11.48)	1.032 (26.21)	.11 (2.8)	.20 (5.1)	.400 (10.16)
U	1 or 2	44	1.20 (30.5) max	1.50 (38.1)	.452 (11.48)	1.281 (32.54)	.16 (4.1)	.215 (5.46)	.500 (12.70)
V	1 or 2	50	1.52 (38.1) max	1.625 (41.28)	.452 (11.48)	1.408 (35.76)	.17 (4.3)	.215 (5.46)	.500 (12.70)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.016 inch (0.41 mm).
4. Metric equivalents are in parentheses.
5. Material: Aluminum alloy in accordance with QQ-A-250/8, temper H32, .032 inch (0.81 mm) thick.
6. Finish: Anodized in accordance with MIL-A-8625 type I or type II.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.
8. Edges of cable entry hole to be rounded to prevent insulation chafing.
9. Shield supplied on connector which also has protective shell must be without aprons.
10. Insert designators are informational and used only when ordering shield/shell as piece parts. Example: M28748/7-Q1001L is a shield for a 14-position insert, with top cable opening, "no polarization, no jackscrews and no contacts".

FIGURE 4. Shield - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, 3, and 4.

Mating connector: See MIL-C-28748/8D.

Part or Identfying Number (PIN): As shown in the following example: 1/

(See (PIN) code)

M28748/7-	A	1	0	L	1A
Basic number of specification sheet	Insert designator (see figure 3)	Shield and shield clamp location (see figure 4)	Shell polarization (see figure 2)	Jackscrews or guidepins (see figure 1)	Contacts

PIN code:

Insert designators	Number of contact positions
A	4
B	7
C	14
D	20
E	26
F	34
G	44
H	50
O 4/	No insert included

Insert designators for piece part orders	Connector sizes
N	4
P	7
Q	14
R	20
S	26
T	34
U	44
V	50

Shield and shield clamp locations	
1	top opening
2	side opening 3/
0	none included

Polarization designators		Polarization positions 2/
A		A
B		B
C	Plug	C
D	shells	D
E		E
F		F
G		G
H	Unpolarized	H
J		A
K		B
L	Receptacle	C
M	shells	D
N		E
P		F
Q		G
R	Unpolarized	H
O	None included	

Jackscrews or guidepins 4/

L - long jackscrews, configuration E, figure 1
F - fixed jackscrews, configuration C, figure 1
S - short jackscrews, configuration D, figure 1
E - miniature jackscrews slotted head, configuration F, figure 1
H - miniature jackscrews hex head, configuration G, figure 1
G - guidepins, configuration B, figure 1
O - none included

Jack and guide location: The jackscrew male (turning or fixed) or guide socket shall be located by the A pin contact.

Contacts: In accordance with MIL-C-28748.
1A - 100 percent size 22
1L - none included 5/

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NOTES:
1/ The number zero (0) is used to indicate which parts are not included.
2/ Polarization positions are for location of plug and receptacle shell pins and slots (see figure 2).
3/ No side clamps available with insert arrangements A, B, D, E, and F.
4/ Manufacturers, mold bodies, and hardware will not necessarily interchange with each other.
5/ These designators are informational and used for insert reference (connector size) to order an appropriate shield or shell as a piece part. Shields and shells may be ordered together or separately.
6/ The "1L" ordering code shall be used in conjunction with insert designator zero "0" or with the insert designators for piece part orders for which no insert is included.

TABLE I. Superseded specification and MS sheets.

Specification or MS sheet	Dated
MIL-C-28748/7C	3 April 1978
MS14012C	28 January 1980
MS14014C	28 January 1980
MS14016C	28 January 1980
MS14018C	28 January 1980
MS14020C	28 January 1980
MS14022C	28 January 1980
MS14024C	28 January 1980
MS14026C	28 January 1980
MS14071C	28 January 1980
MS14072C	28 January 1980

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV, ME
Navy - MC, OS

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5935-3881-01)

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